Constraining Metastable Very Heavy Dark Matter using the Extragalactic Gamma-Ray Background

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Abstract

There has been extensive work in constraining the properties of dark matter particles through astronomical observations and multi-messenger astrophysics experiments. In this work, we constrain the mass and lifetime of metastable very heavy dark matter (VHDM) particles using blazar emissions and the extragalactic gamma-ray background (EGB) as measured by the Large Area Telescope onboard the *Fermi Gamma-ray Space Telescope*. We synthesize a model of decaying dark matter particles that incorporates astrophysical and particle physics contributions to the gamma-ray intensity. The former involves calculating the average $\mathcal{J}$ factor along lines of sight using the Navarro-Frenk-White density profile, and the latter consists of using the HDMSpectra tool to calculate gamma-ray spectra from dark matter decay channels. We place limits on dark matter models that when added to blazar emission overproduce the EGB on a 2σ level. Our results suggest that if VHDM particles contribute to the EGB, they must have mean lifetimes in excess of 10^{25} seconds.

1 Introduction

A variety of diverse astronomical observations indicate that there may be some substance that interacts gravitationally, but not electromagnetically. Known as dark matter, this mysterious substance is estimated to account for nearly a quarter of the universe's energy content. While greatly outnumbering normal matter, dark matter has not been directly detected, nor is there consensus on its precise composition or behavior.

In the early 17th century, astronomer Johannes Kepler formulated laws of planetary motion based on observations and empirical data he collected of celestial objects in the solar system. Later in the century, Isaac Newton's laws of motion and law of universal gravitation demonstrated that Kepler's laws would apply to any system of objects interacting gravitationally. Kepler's third law of planetary motion describes the inverse relationship between an orbiting object's semi-major axis and its orbital period. While this dependence holds for sub-galactic scale systems, the third law cannot model behavior observed in the rotation curves of disc galaxies using solely visible matter [1]. This problem of missing mass has also been observed in the behavior of stellar velocity dispersions, galaxy clusters, and in the cosmic microwave background (CMB). Dark matter is the prevailing explanation for these discrepancies in mass.

Galaxy clusters provide a plethora of observational evidence for the existence of dark matter. Gravitational lensing, the phenomenon where massive objects between a source and an observer can bend light as described by Albert Einstein's theory of general relativity, is typically observed around clusters of galaxies. Gravitational lensing caused by massive objects produces distortions in and duplicates of images of background objects. Analyzing the geometry of distortions in images and the mass-to-light ratios of galaxy clusters allows for the determinations of dark matter distributions within galaxies [2]. These kinds of measurements have found that dark matter is the major constituent of galaxy clusters [3]. Perhaps the most notable observational evidence for dark matter is the Bullet Cluster, a structure consisting of two colliding clusters of galaxies. When viewed in the visible band, the relationship between the two clusters is not clear. When observed in the x-ray band however, the Bullet Cluster is characterized by regions of super-heated gas as well as a shock wave formed from the collision. Obtaining the distributions of the clusters' masses through gravitational lensing has uncovered that the mass distribution does not coincide with the gas distribution, suggesting that much of the clusters' masses did not interact during the collision. These observations of the Bullet Cluster provide serious challenges for alternative gravity theories, and provide the strongest evidence for the existence of dark matter [4].

Not only is dark matter's influence measurable in localized structures in the universe, but its effects can also be seen permeating the universe. The CMB, leftover radiation from the Big Bang, has been imprinted on differently by the perturbations of dark matter than by those of baryonic matter. Very subtle temperature fluctuations in the CMB are thought to have generated the filamentary structure of the universe on the largest scales, and modeling these fluctuations necessitates the inclusion of dark matter. In addition, measurements of the CMB reinforce evidence of dark matter being the dominant form of matter [5]. Furthermore, the imbalance of matter and antimatter in the universe may point to evidence for dark matter [6]. Decaying or annihilating dark matter particles could potentially generate electrons and positrons in accordance with measured spectra.

While the consensus is that dark matter is non-baryonic, there are many hypotheses about the physical form of dark matter. These dark matter candidates include supersymmetric particle counterparts, axions, sterile neutrinos, macroscopic compact halo objects (MaCHOs), and weakly interacting massive particles (WIMPs). The WIMP model of dark matter had been especially intriguing as these families of particles are potentially detectable in accelerator experiments [7]. The so far non-detection of WIMPs implies that if they exist, their energies are in excess of current experimental limits. Furthermore, the WIMP model has been analyzed extensively in context of astroparticle observations. This constrains weakly interacting dark matter particles to be very massive, known as very heavy dark matter (VHDM) [8]. Additionally, dark matter particles may be stable or unstable, as long as their lifetime is longer than the age of the universe (metastable). For generic annihilating or decaying dark matter candidates, the search for VHDM may naturally extend into the realm of multimessenger astronomy in the observation of the spectra of stable final-state particles.

Dark matter particles may follow various allowed decay or annihilation channels, spawning cosmic rays such as neutrinos or gamma-rays [9, 10]. Neutrinos serve as astrophysical probes into the early and distant universe, and may provide great insight into the Standard Model of particle physics if they are connected to dark matter. Gamma-rays are produced in a myriad of high-energy reactions, and instruments like the Large Area Telescope onboard the *Fermi Gamma-ray Space Telescope (Fermi)* have identified various gamma-ray sources [11, 12, 13, 14]. However, there also exists an unresolved background of gamma-rays, known as the isotropic diffuse gamma-ray background (IGRB). The IGRB as measured by *Fermi* spans from 100 MeV to 820 GeV, and is made of contributions that are believed to mostly originate outside of the Milky Way galaxy. The majority of the IGRB is attributed to an unresolved component in the extragalactic gamma-ray background (EGB), potentially along with a largely isotropic Galactic halo emission, both being subject to systematic uncertainties due to diffuse Galactic (foreground) gamma-rays. The EGB can be modeled as a power law over almost four orders of magnitude in energy [15], and efforts have been made to model it using a variety of astronomical emissions (ex. blazars, star-forming regions, etc.) [16]. The potential contribution of decaying dark matter to the EGB is yet to be fully understood [17], and is explored in this work.

Previous studies have employed analyses of constraining dark matter properties using the IGRB [8, 16, 18, 19, 20, 21, 22, 23], other gamma-ray observations [24, 25, 26, 27], x-rays [28, 29, 30], other cosmic rays [31, 32], and neutrinos [8, 33, 34, 35]. In this work, we constrain the mass and lifetime of unstable VHDM particles by evaluating the contribution of Galactic dark matter and extragalactic blazars to the EGB. This is accomplished by fitting models of gamma-ray intensities generated by blazars and decaying dark matter to the EGB spectrum by minimizing a χ^2 test statistic. Dark matter particle property constraints are shown as a plot of the minimum dark matter lifetime as a function of dark matter mass. Minimum lifetimes are imposed by ruling out dark matter models that when added to blazar emission overproduce the EGB emission at a 2σ level.

The organization of this paper is as follows. In Section 2, we first provide an overview of calculating the spectra of particles created by decaying dark matter. Then, we provide an overview of calculating the Galactic contribution of dark matter. In Section 3, we discuss the blazar component of flux that is used as the extragalactic contribution in this work. In Section 4, we describe the EGB data, the fitting test statistic, and the methodology used to generate constraints. The constraints themselves are shown in Section 5 and are discussed in Section 6.

2 Galactic Contribution

The intermediate states resulting from dark matter decay are unknown. As representative channels, we consider gauge bosons (W^+W^-), heavy quarks ($b\bar{b}$), heavy leptons ($\tau^+\tau^-$), and neutrinos ($\nu\bar{\nu}$). The spectra of secondary particles resulting from these channels are calculated using HDMSpectra [36]. HDMSpectra is a user-friendly Python library used for generating spectra of particles originating from dark matter decay and annihilation from the TeV to Planck scale. HDMSpectra builds upon physics from similar tools such as PYTHIA or PPPC4DMID with improvements across the mass scale of the progenitor dark matter particle. Within HDMSpectra, non-dimensionalized, differential, prompt spectra are generated as a function of the fraction of dark matter mass. That is, the spectra of final state particles S generated by decaying dark matter are defined as

$$\frac{dN_S}{dx}(x) \text{ for } x = \frac{2E}{m_{\text{dm}}}, \tag{1}$$

where E is the energy of the final state particle, and m_{dm} is the assumed dark matter particle mass in units of GeV. For a detailed analysis of the EGB, the contribution of cascading electrons and positrons as well as propagation effects must be included. In this project, we only focus the prompt spectrum of gamma-rays from decaying dark matter. An example of the gamma-ray spectrum from decaying dark matter is shown in Figure 1 for the representative decay channels.

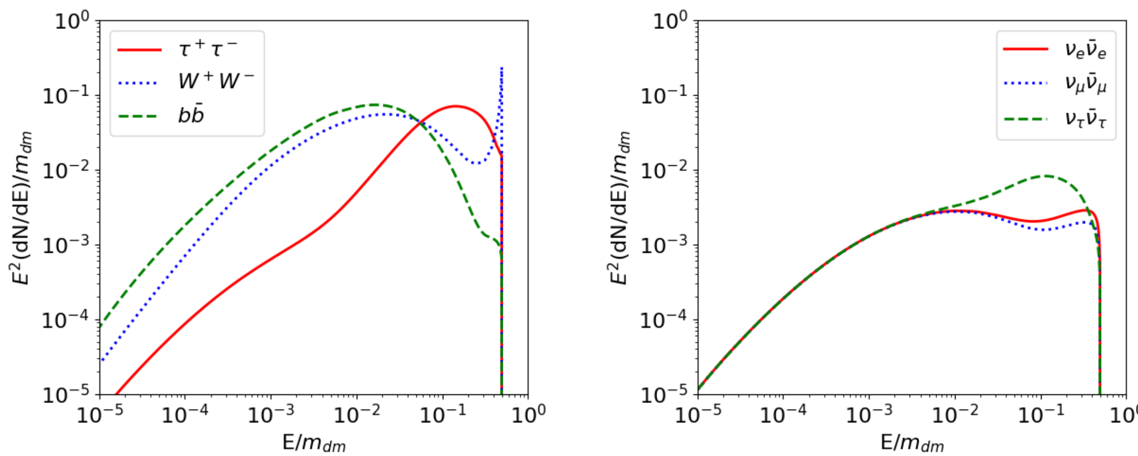


Figure 1: Prompt gamma-ray spectra generated in HDMSpectra for DM $\rightarrow W^+W^- \rightarrow \gamma$, DM $\rightarrow b\bar{b} \rightarrow \gamma$, and DM $\rightarrow \tau^+\tau^- \rightarrow \gamma$ decay channels (left) as well as from electroweak bremsstrahlung for e, μ, and τ neutrino flavors (right) with m_{dm} = 100 TeV. Note that the spectra in these plots are dN/dE. The behavior of spectra arise from complex particle physics interactions based on the current understanding and computational implementation of the Standard Model in HDMSpectra.

If decaying dark matter contributes to the IGRB, it is necessary to quantify the amount of dark matter within the Galaxy. Simulations have demonstrated that dark matter in galaxies is organized in approximately spherically symmetric halos around galactic centers. Several dark matter density profiles have had success in replicating behavior seen in simulations, such as the Navarro-Frenk-White (NFW) profile used in this work, given as

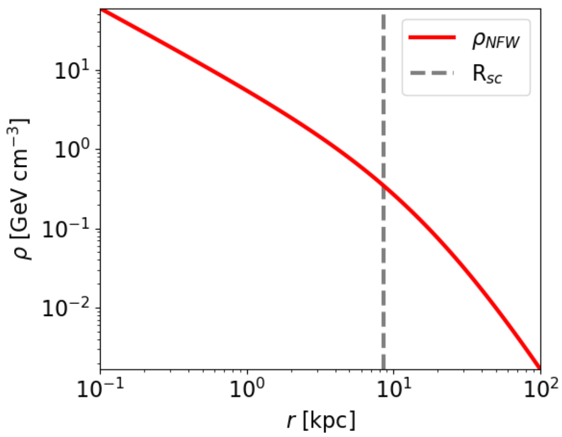


Figure 2: Normalized NFW dark matter energy density profile with the radius of the solar circle labeled.

$$\frac{\rho(r)}{\rho_0} = \frac{1}{(r/r_s)(1 + r/r_s)^2}. \tag{2}$$

Equation 2 describes the behavior of non-dimensionalized dark matter density as a function of radius r from the Galactic Center (GC), where ρ_0 is a scale density and r_s is a scale radius [37]. We normalize the NFW profile with $r_s = R_{\text{mw}}$ = 20 kpc as the radius of the Milky Way galaxy and $\rho_0 = \rho_{\text{sc}}$ = 0.3 GeV cm^{-3} as the local energy density at the solar circle [8]. This normalization is shown in Figure 2.

The IGRB is isotropic – independent of the direction of observation. As such, quantifying the contribution of decaying dark matter to the IGRB involves calculating how much dark matter is encountered along a line of sight for a given observation direction, then averaging across all directions. More formally, we use the $\mathcal{J}$ factor, a dimensionless quantity accounting for the column density of dark matter along a line of sight at an angle ψ with respect to the GC direction. For decaying dark matter, the $\mathcal{J}$ factor is given as

$$\mathcal{J}^{dec}(\psi) = \frac{1}{R_{\text{sc}}\rho_{\text{sc}}} \int_0^{l_{\max}(\psi)} \rho_{\text{dm}}(r(\psi, l))\, dl, \tag{3}$$

where $l_{\max} = R_{\text{sc}} \cos(\psi) + \sqrt{R_{\text{mw}}^2 - R_{\text{sc}}^2 \sin^2(\psi)}$ is the distance in kpc to the end of a line of sight, l is a distance along a line of sight, $r = \sqrt{R_{\text{sc}}^2 + l^2 - 2lR_{\text{sc}} \cos(\psi)}$, R_{sc} = 8.5 kpc as the radius of the solar circle, and ρ_{dm} is the NFW profile [8]. The angle ψ, the pointing angle of a line of sight with respect to the GC direction, can be related to Galactic coordinates ℓ and b via the relation

$$\psi = \arccos(\cos(\ell)\cos(b)). \tag{4}$$

A derivation of the above geometric quantities can be found in the Appendix. To evaluate Equation 3, the NFW density profile we use is given by

$$\rho_{\text{dm}}(r) = \frac{\rho_{\text{sc}}}{(r/R_{\text{mw}})(1 + r/R_{\text{mw}})^2}, \tag{5}$$

but is adjusted to have a flat core to alleviate issues with numerical integration along lines of sight that pass near the GC [38]. The revised density profile, given by

$$\rho_{\text{dm}}(r) = \begin{cases} \frac{\rho_{\text{sc}}}{(0.015\ \text{kpc}/R_{\text{mw}})(1+0.015\ \text{kpc}/R_{\text{mw}})^2} & \text{if } r < 0.015\ \text{kpc} \\ \frac{\rho_{\text{sc}}}{(r/R_{\text{mw}})(1+r/R_{\text{mw}})^2} & \text{if } r \geq 0.015\ \text{kpc} \end{cases}, \tag{6}$$

most notably affects lines of sight within $0 \leq \psi \leq 0.1^\circ$ of the GC, and does not significantly impact results.

It is also useful to calculate $\mathcal{J}$ within a field of view. Since $\mathcal{J}$ is maximized in the direction of the GC ($\psi = 0$), we center a field of view on the GC. The integral of $\mathcal{J}$ within such a field of view of half angle ψ is labeled as $\mathcal{J}_\psi$ and is defined as

$$\mathcal{J}_\psi = \int_0^\psi \mathcal{J}(\psi') \sin(\psi')\, d\psi', \tag{7}$$

where ψ' is a dummy integration variable. If $\mathcal{J}_\psi$ is then divided by the integral of $\mathcal{J}$ over the whole sky ($\mathcal{J}_{180^\circ}$), the fraction of total dark matter contained within a field of view can be found. The result of this calculation is shown in Figure 3, demonstrating that 25% and 50% of dark matter can be found within fields of view with half-angles approximately 33° and 58° respectively. Using Equation 3, we generated a skymap of $\mathcal{J}$ factors and plotted the 25% and 50% dark matter contours as shown in Figure 4. These contours will be used in a future analysis to involve the directional distribution of gamma-rays.

Finally, the average of $\mathcal{J}$ within a field of view can be found by dividing Equation 7 by a factor proportional to the solid angle. This quantity is given as

$$\mathcal{J}_\Omega = \frac{1}{1 - \cos(\psi)} \int_0^\psi \mathcal{J}(\psi') \sin(\psi')\, d\psi', \tag{8}$$

and is shown on the right of Figure 3. To model the contribution of dark matter decays to the EGB, $\mathcal{J}_\Omega(180^\circ)$ is used as the proportionality constant for the isotropic level of Galactic dark matter.

The directional Galactic contribution prompt flux (in units of particles GeV^{-1} cm^{-2} s^{-1} sr^{-1}) for decaying dark matter is given by

$$\Phi^{\text{dec}}(\psi) = \frac{R_{\text{sc}}\rho_{\text{sc}}}{4\pi m_{\text{dm}}\tau_{\text{dm}}} \frac{dN_S}{dE} \mathcal{J}^{\text{dec}}(\psi), \tag{9}$$

where τ_{dm} is the assumed mean lifetime of dark matter particles and dN_S/dE is the prompt spectrum of a dark matter decay product [8]. In general, the mean lifetime τ is a parameter in the differential equation for exponential decay of some quantity n with the form

$$\frac{dn}{dt} = -\frac{n}{\tau} \tag{10}$$

as a function of time t. Here, τ_{dm} is the time after which the population of dark matter particles is reduced to $1/e$ times its initial value, where e is Euler's number.

For the purpose of modeling the EGB, Equation 9 is modified to encode the isotropic gamma-ray flux. Equation 11 is the complete dark matter component of the EGB model for this work, where dN$_\gamma$/dE is the photon spectrum for a particular decay channel.

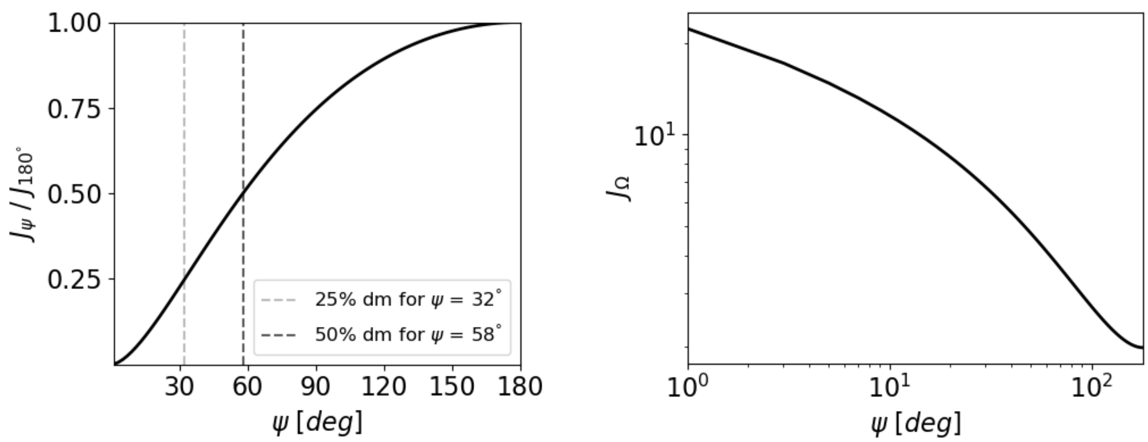


Figure 3: Left: The fraction of dark matter contained within a field of view of half angle ψ calculated using Equation 7. Right: The average value of $\mathcal{J}$ within a field of view of half angle ψ calculated using Equation 8.

$$\Phi_{\rm iso}^{\rm dec} = \frac{R_{\rm sc}\rho_{\rm sc}}{4\pi m_{\rm dm}\tau_{\rm dm}}\frac{dN_\gamma}{dE}\mathcal{J}_\Omega(180^\circ) \tag{11}$$

This is the prompt flux of gamma-rays generated by decaying VHDM, hereafter referred to as the dark matter intensity.

3 Isotropic Contribution

Both Galactic and extragalactic components contribute gamma-rays to the IGRB. In this work, we represent the Galactic contribution as the dark matter intensity. For the extragalactic contribution, we use the spectrum of the integrated emission of blazars from Ref. [16].

An Active Galactic Nucleus (AGN) is a compact region at the centers of galaxies with a very high luminosity. Usually, these high luminosities are in portions of the electromagnetic spectrum not associated with radiation produced by stars. The prevailing explanation of this light is supermassive black holes at the centers of galaxies which radiate a portion of energy gained from accreting matter. AGNs are usually characterized by two multi-light-year-long relativistic jets protruding out of opposite sides of the AGN's central compact object. These central objects include black holes, neutron stars, or white dwarfs. The most energetic AGNs are called quasars (also known as quasi-stellar objects), and AGNs whose jets are directed very nearly at Earth are called blazars.

In addition to the IGRB, *Fermi* has detected many astronomical sources of gamma-rays including blazars [39]. Among astronomical classes of objects detected by *Fermi*, blazars constitute the largest population of gamma-ray sources, and may contribute to around 50% of EGB photons [16]. Furthermore, Ref. [16] has derived new blazar luminosity constraints to produce a realistic model of the spectra of blazar emissions. This makes blazars adequate extragalactic gamma-ray sources for our purposes of generating constraints of dark matter particle masses and lifetimes.

The differential blazar spectrum was extracted from Figure 3 of Ref. [16] along with its uncer-

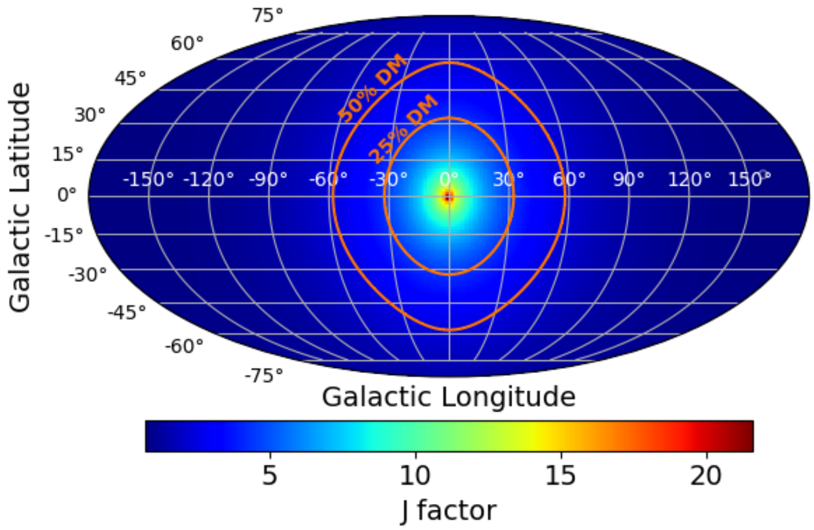


Figure 4: Mollweide projection of $\mathcal{J}$ factor values across the sky as a function of Galactic Longitude ℓ and Latitude b. This plot shows differential $\mathcal{J}$ factors in the sense that a particular line of sight is represented by an infinitesimal area.

tainties[1]. Symmetric uncertainties on the blazar spectrum were calculated by extracting the lower and upper bounds of the blazar intensity from the same figure and calculating their mean differences from the central values. Total blazar intensities and uncertainties were obtained by integrating over energy bins.

4 Methods

The *Fermi* collaboration gives the EGB spectrum in terms of binned total EGB intensity (flux in units of photons cm^{-2} s^{-1} sr^{-1}) in Table 3 of Ref. [15], along with background and foreground uncertainties[2]. It is often more meaningful to plot the differential energy flux (in units of GeV cm^{-2} s^{-1} sr^{-1}), and we will do so in this paper where relevant. In our methodology however, we always use total intensities. The differential energy flux is obtained from the total intensity by multiplying the binned total intensity by a factor $E^2/E_{\text{bin size}}$. A plot of the EGB and blazar spectra and their uncertainties are shown in Figure 5.

A starting point for placing constraints on dark matter particle properties is considering the dark matter mass range to explore. HDMSpectra can generate spectra with dark matter masses from 10^3 to 10^{19} GeV. However, the spectra of final state particles are only defined for energy fractions within $10^{-6} \leq$ x ≤ 1 for x in Equation 1. Therefore, if we require that the dark matter decay gamma-ray spectrum be defined over the entire EGB (0.1 - 820 GeV) in accordance with the limits of HDMSpectra, we find the range of dark matter masses of 1.7×10^3 GeV $\leq m_{\text{dm}} \leq 2 \times 10^5$ GeV to consider. Figure 6 shows the dark matter intensity (Equation 11) for this mass range and a single value of τ_{dm} across the representative decay channels.

[1] WebPlotDigitizer is a useful tool for extracting data points from digital images and can be found at `https://apps.automeris.io/wpd/`.
[2] Table 3 can be found at `https://iopscience.iop.org/article/10.1088/0004-637X/799/1/86#apj504089t3`

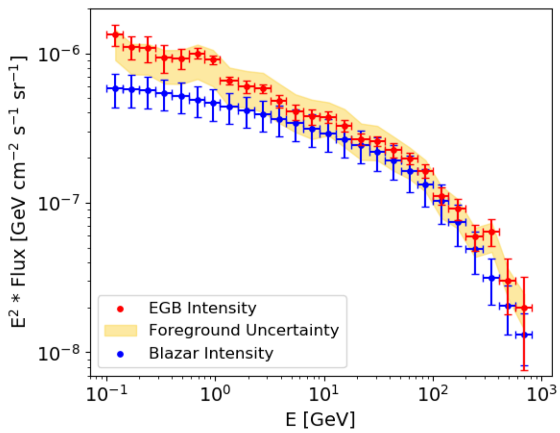


Figure 5: Differential EGB energy flux as calculated from Table 3 of Ref. [15]. Error bars on the x-axis denote the sizes of energy bins. For the EGB intensities, error bars on the y-axis denote the statistical and systematic uncertainties of the EGB intensity added in quadrature. For the blazar intensities, y-axis errors denote uncertainties from Figure 3 of Ref. [16].

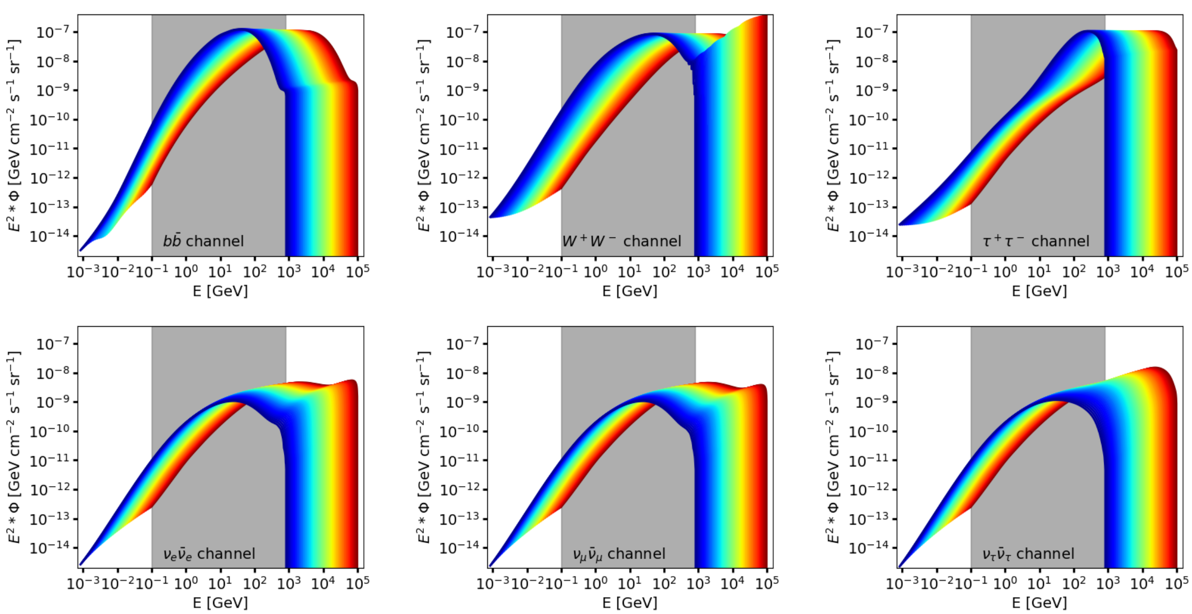


Figure 6: Differential gamma-ray energy spectra from decaying dark matter particles as defined in Equation 11 across multiple channels for the dark matter mass range of 1.7×10^3 GeV $\leq m_{\text{dm}} \leq 2 \times 10^5$ GeV and $\tau_{\text{dm}} = 10^{27}$ GeV. Warmer colors indicate higher dark matter masses, cooler colors indicate lower. The gray band indicates the energy range that the EGB is defined on.

To generate the constraints themselves, we use Bayesian statistical inference. When fitting a model to data, it is often most natural to use a generative probabilistic model, a statistical model of the joint probability distribution between an observable variable and a target variable. With such a model, one can generate random samples (observable variables) as functions of model parameters to compare against data (target variables) in order to find sets of parameters which fit the model well. This involves defining a likelihood function, a measure of the goodness of a fit of a statistical model to data for certain values of unknown parameters. Maximizing the likelihood function is known as maximum likelihood estimation (MLE), and provides estimates of model parameters that have the highest likelihood of producing data. We generate constraints on dark matter mass and lifetime using MLE and following the methodology discussed in Ref. [16]. It is most convenient to deal with the natural logarithm of the likelihood function, referred to as the log likelihood. We start by defining the log likelihood function in accordance with Ref. [16] as

$$\chi^2_{m_{\rm dm}}(\tau_{\rm dm}, \mathscr{A}) = \sum_i^N \left(\frac{(F_{i,EGB} - \mathscr{A} F_{i,Blazar} - \Phi_{i,tot\ \rm dm}(m_{\rm dm}, \tau_{\rm dm}))^2}{\sigma_i^2} + \frac{(1-\mathscr{A})^2}{\sigma^2_{\mathscr{A}}} \right), \tag{12}$$

where the sum is evaluated over the N energy bins of the EGB spectrum. $F_{i,EGB}$ and $F_{i,Blazar}$ are the total EGB and blazar intensities, $\Phi_{i,tot\ \rm dm}(m_{\rm dm}, \tau_{\rm dm})$ is the total dark matter intensity (Equation 11 integrated over the EGB spectrum energy bins), $\mathscr{A}$ is the blazar intensity renormalization constant, a nuisance parameter used to account for additional astronomical contributions not explicitly included in this analysis, and $\sigma_{\mathscr{A}} = \langle \sigma_{i,Blazar}/F_{i,Blazar} \rangle$ is its average uncertainty. Lastly, $\sigma_i^2 = \sigma^2_{i,EGB} + \sigma^2_{i,FG}$ is the sum in quadrature of the total EGB and foreground uncertainties.

For a given $m_{\rm dm}$, we find optimal values of $\tau_{\rm dm}$ and $\mathscr{A}$ that minimize Equation 12 using Markov Chain Monte Carlo (MCMC). MCMC is a class of statistical analysis algorithms used to find model parameters from known data. We use emcee[3], a Python based implementation of MCMC [40]. To start, emcee draws parameter samples from a log posterior probability function given by the left-hand side of Equation 13,

$$p(\tau_{\rm dm}, \mathscr{A} | E, F_{EGB}, \varsigma) \propto p(\tau_{\rm dm}, \mathscr{A}) + p(F_{EGB} | E, \varsigma, \tau_{\rm dm}, \mathscr{A}), \tag{13}$$

where ς denotes the collection of uncertainties in EGB and blazar values. The rightmost term is the log likelihood function, as given in Equation 12:

$$p(F_{EGB} | E, \varsigma, \tau_{\rm dm}, \mathscr{A}) = \chi^2_{m_{\rm dm}}(\tau_{\rm dm}, \mathscr{A}) \tag{14}$$

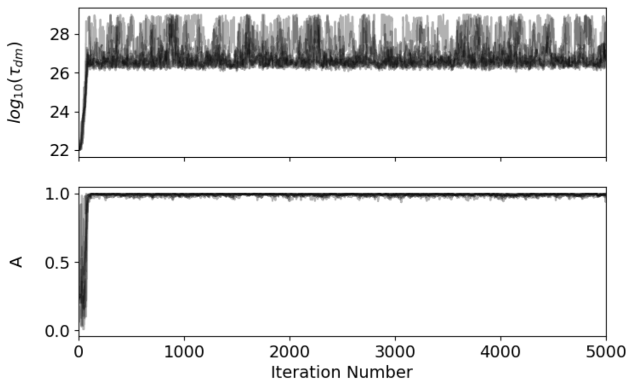
The remaining term is the log prior function, which encodes prior knowledge of the parameters such as the valid regions of the parameter space. We define our log prior as

$$p(\tau_{\rm dm}, \mathscr{A}) = \begin{cases} 0 & \text{if } \tau_{\rm dm} > t_0 \text{ and } 0 \leq \mathscr{A} \leq 1 \\ -\infty & \text{otherwise} \end{cases}, \tag{15}$$

where $t_0 = 4.343 \times 10^{27}$ seconds, the current age of the universe [5]. If emcee samples values of $\tau_{\rm dm}$ and $\mathscr{A}$ outside of the valid parameter space defined in the prior, the log posterior probability function takes its lowest value[4]. Along with Equation 13, emcee initializes a number of "walkers" within the parameter space. These walkers are linearly independent ensemble samplers which "walk" throughout the parameter space in order to evaluate the log posterior probability multiple times per iteration. Using more walkers may lessen the number of iterations needed for emcee to

[3]emcee, the MCMC Hammer, can be found at `https://emcee.readthedocs.io/en/stable/`.

[4]In practice, the log posterior probability (more precisely, the log likelihood) is not sensitive to large changes in values of $\tau_{\rm dm}$ in regions of the parameter space where $\mathscr{A} \approx 1$ and $\tau_{\rm dm} >> t_0$. This causes the emcee walkers to get stuck in a region of the parameter space where $\tau_{\rm dm} >> 10^{100}$ seconds. This problem was alleviated by adding an upper bound to the $\tau_{\rm dm}$ prior such that there is minimal degeneracy in the log likelihood function.



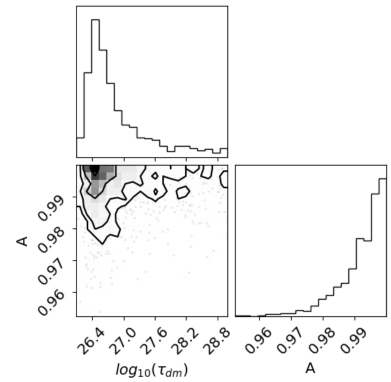


Figure 7: Walker positions as a function of the iteration number (left) and the corner plot (right) for the $\tau^+\tau^-$ channel and $m_{\rm dm} = 1 \times 10^4$ GeV. Within the walker position subplots, the heaviness of the shading denotes the location of the majority of the walkers; heavily shaded regions show where most of the walkers are in the parameter space. For the corner plot, the marginalized distribution of the likelihoods of values for each parameter independently is shown on the diagonal, and the marginalized two dimensional distribution is shown in the remaining panel.

converge, but will increase the computation time per iteration. In this work, we use 8 walkers and let each chain run for 5,000 iterations or until emcee converges. Convergence is reached when the autocorrelation times for each parameter is greater than $\mathscr{N}/50$, where $\mathscr{N}$ is the length of the chain. We show an example of the positions of walkers in the parameter space as a function of the chain length (iteration number) along with a corner plot in Figure 7. The corner plot shows the one and two dimensional projections of the posterior probability distributions of $\tau_{\rm dm}$ and $\mathscr{A}$. Corner plots are useful in general to qualitatively analyze the covariances between parameters.

In this work, we constrain dark matter models that, together with blazar emission, overproduce the EGB intensity at a 2σ level. The schema to find the minimum dark matter lifetime is as follows, and can be followed along graphically with an example in Figure 8. For a single dark matter mass, we set $\mathscr{A}$ in Equation 12 equal to its optimal value. In practice, $\mathscr{A} \approx 1$ for all sampled masses and decay channels studied, and was set equal to 1 across all values of dark matter mass for ease of calculation. As such, $\chi^2_{m_{\rm dm}}$ (blue curve) is a function of one variable: $\tau_{\rm dm}$. In accordance with χ^2 statistics, we obtain the aforementioned 2σ dark matter model limit (dashed red line) by finding the value of $\tau_{\rm dm}$ that worsens the value of the optimized $\chi^2_{m_{\rm dm}}$ (dashed purple line) by ≥ 4 (dashed turquoise line). Values of $\tau_{\rm dm}$ below the 2σ limit are ruled out (shaded gray region). This method was repeated for multiple dark matter masses from 1.7×10^3 to 2×10^5 GeV, and was used to generate the constraints found in Section 5. In Figure 9, we show examples of the total dark matter intensity added to the total blazar intensity along with how the sum compares to the EGB for a single value of $m_{\rm dm}$ and $\tau_{\rm dm}$.

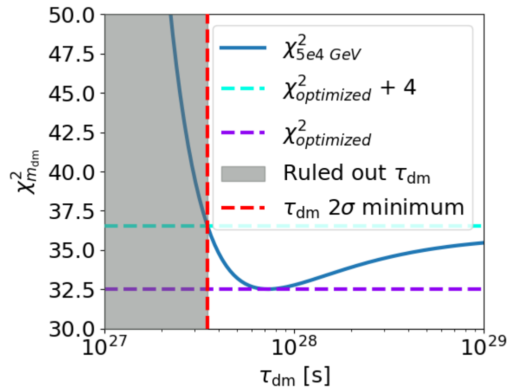


Figure 8: Schema of the method used to find dark matter lifetime and mass constraints.

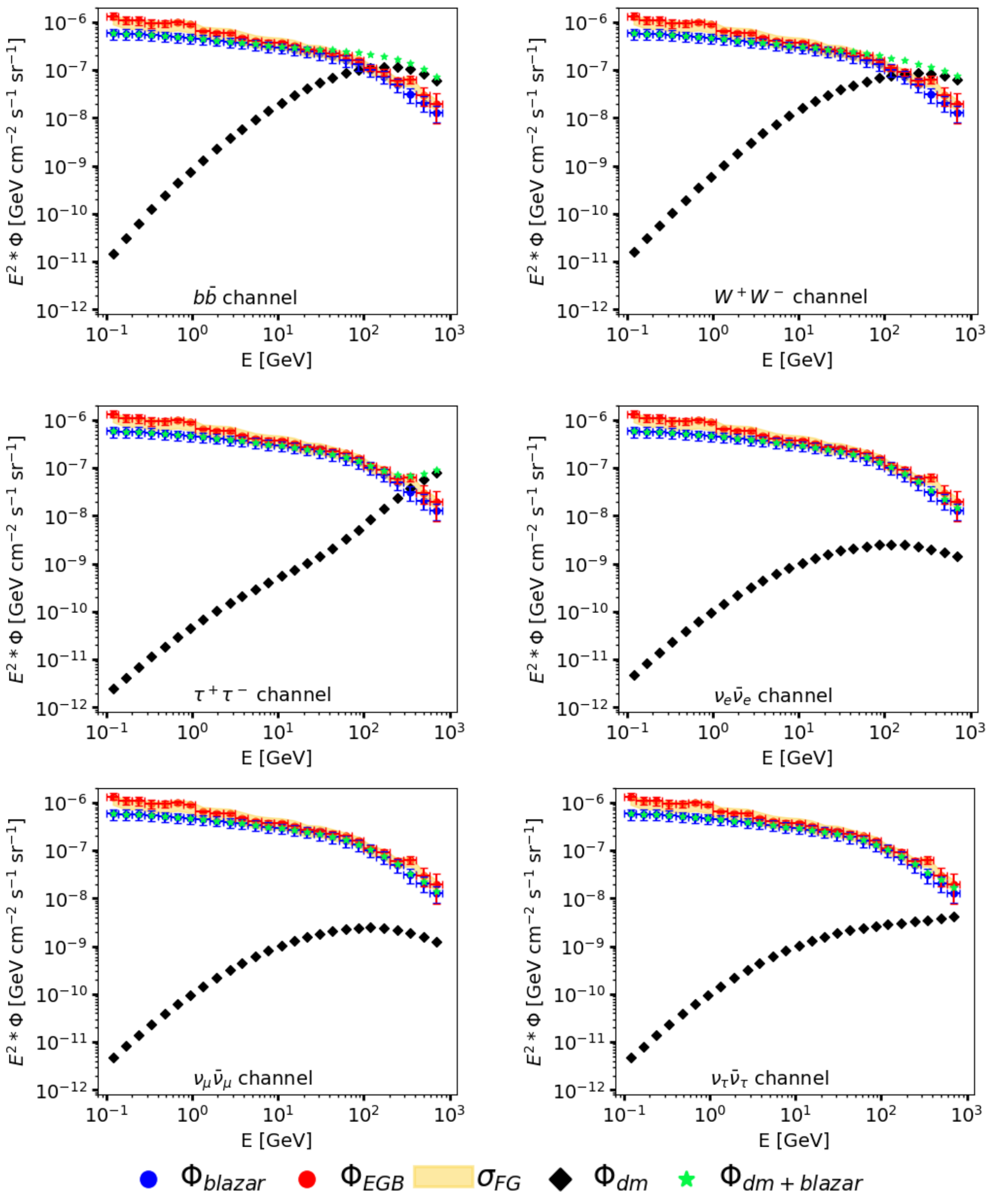


Figure 9: Plots of differential dark matter and blazar energy spectra as well as their sum compared to the EGB spectrum for $m_{\rm dm} = 1 \times 10^4$ GeV and $\tau_{\rm dm} = 10^{27}$ s across the representative decay channels.

It is clear that the blazar intensity approximately matches the shape of the EGB spectra across all measurement bins. From around 10 - 820 GeV the blazar intensity closely matches the magnitude of the EGB, leaving little room for dark matter signals. For the lower energy EGB bins between 0.1 - 10 GeV, lower mass dark matter could potentially contribute a larger signal.

5 Results

In this section, we show our constraints on dark matter properties derived from fitting the dark matter intensity plus blazar model to the EGB spectrum. The constraints consist of plots of minimum dark matter lifetimes as a function of mass for each decay channel studied. All constraints are shown in Figure 10.

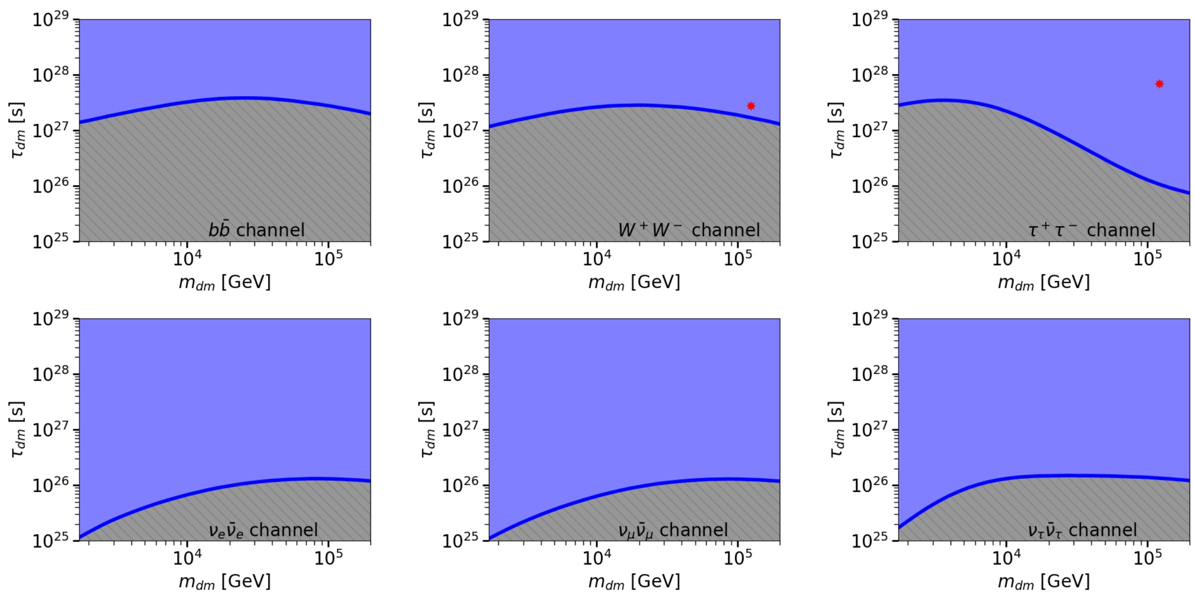


Figure 10: VHDM particle lifetime and mass constraints. Each subplot shows the constraints for a specific decay channel. As described in Section 4, the thick blue curves shows the 2σ minimum lifetimes, the hashed gray regions (extending down to zero) denote the parameter space of invalid dark matter particle properties ruled out by EGB intensities, and the shaded blue regions (extending infinitely upwards) denote valid dark matter particle properties not ruled out. Red stars denote the best fit parameters for modeling the IceCube flux as given in Ref. [24].

Following from Figure 10, all constraints are characterized by a concave-down behavior. The most stringent constraints are in the W^+W^- and $b\bar{b}$ channels, where the minimum lifetimes must be in excess of at least one billion times longer than the current age of the universe. There is more variance in the constraints for the $\tau^+\tau^-$ channel, as well as an apparent change in concavity occurring around $m_{\rm dm} = 3 \times 10^4$ GeV. The electroweak bremsstrahlung channels give weaker constraints, placing limits on the minimum dark matter lifetimes that are at least one order of magnitude lower than the other three channels.

The long lifetime of VHDM particles as suggested by this work may be a possible explanation for diffuse neutrinos detected by IceCube [33]. The best fit values of dark matter particle mass and lifetime for modeling IceCube flux for the W^+W^- and $\tau^+\tau^-$ channels as derived in Ref. [24] are within the valid parameter space constrained by this work, as seen in Figure 10. The consistency between our constraints and the best fit parameters derived in Ref. [24] allows for decaying VHDM particles to contribute to the flux of astrophysical neutrinos.

6 Summary and Discussion

In this paper, we explored the potential contribution of metastable very heavy dark matter to the *Fermi* extragalactic gamma-ray background data along with blazars and derive constraints on the lifetime and masses of decaying dark matter particles. Using the Navarro-Frenk-White density profile for Galactic dark matter, we calculate $\mathcal{J}$ factors across lines of sight then compute the angular average to obtain the isotropic level of dark matter in the Milky Way. We use the HDMSpectra tool to generate the spectra of gamma-rays associated with final state particles such as heavy leptons, heavy quarks, and guage bosons, as well as electroweak brehmstrallung from all three neutrino flavors. We combined the astrophysical and particle physics components to obtain a model of gamma-rays generated by decaying very heavy dark matter particles. The blazar contribution to the EGB was modeled using the theoretical blazar intensities in Ref. [16]. We explored two orders of magnitude in dark matter mass such that all dark matter model spectra are defined over the range of the EGB using HDMSpectra. Constraints were derived by defining a χ^2 test statistic explicitly involving EGB, blazar, and Galactic foreground uncertainties as well as a free blazar renormalization parameter. We used Markov Chain Monte Carlo to optimize the test statistic to find relationships between dark matter particle mass and the minimum lifetime for models that when added to the blazar intensity, overproduce the EGB on a 2σ level. In each decay channel studied, we found that the minimum dark matter lifetime must be in excess of 10^{25} seconds and exhibits a mostly concave down behavior. Two decay channels that place stronger constraints on dark matter lifetime around 10^{27} seconds extend the plausibility of decaying dark matter's contribution to astrophysical neutrinos detected by IceCube.

The constraints derived in this work improve upon previous studies by making use of HDMSpectra's greater accuracy in implementing Standard Model interactions. Full treatments of electromagnetic cascades, inverse Compton scattering, and $\gamma\gamma$ interactions with the extragalactic background light/CMB are needed to account for gamma-ray propagation effects and other phenomena. While blazars contribute to a large fraction of EGB photons, including additional astronomical sources may allow for imposing stricter constraints on very heavy dark matter particle properties. These may include radio galaxies and star forming galaxies as used in Ref. [16].

The constraints in this work are consistent with those derived in Ref. [8] for the $b\bar{b}$ and W^+W^- channels, roughly matching the order of magnitude of those constraints for the dark matter particle mass range explored here. Discrepancies in these constraints likely lies in part to the greater accuracy afforded by HDMSpectra over PYTHIA as well as the cascades, propagation effects, and the contribution of extragalactic dark matter neglected in this work. The concavity of the constraints in this work are qualitatively similar to the non-neutrino channel constraints derived in Ref. [18], again differing roughly by an order of magnitude. Along those channels, our constraints are in nearby orders of magnitude of those derived in Refs. [19], [20], [23], and [24], and have similar behaviors regarding concavity. Deriving constraints on the self-annihilation cross section of VHDM particles and comparing with other works can shed further light on the feasibility of our constraints.

Heavier dark matter models can be analyzed by extrapolating the spectra for energy fractions below 10^{-6}. Additional sets of constraints can be generated by analyzing the contribution of dark matter to neutrino flux. The connections between high-energy gamma-rays and neutrinos has significant consequences for multi-messenger astronomy, and VHDM dark matter particles may provide additional insight into understanding various astrophysical phenomena. This avenue of analysis may uncover insights into high energy neutrino events detected by IceCube. Recreating a similar analysis done in Ref. [24] to derive more best fit parameters to model IceCube neutrino flux will provide additional reasoning for the feasibility of our constraints in the context of astrophysical neutrinos. Along with neutrinos and gamma-rays, other astrophysical messengers can be studied to produce robust multi-messenger constraints, such as electrons and positrons in the context of the positron excess and the AMS-02 experiment. Finally, it is trivial to reproduce the methodology in this work to derive constraints for the full suite of decay channels included in HDMSpectra.

In the next stage of this work, we will include corrections related to cascades and gamma-ray propagation effects and recalculate our constraints for decaying VHDM. Following the methodology in Refs. [16] and [8], it is straightforward to augment our analysis for annihilating VHDM particles and generate new constraints. We also look to improve our constraints by calculating the contribution of extragalactic dark matter particles to the EGB, as well as involve angular information about gamma-rays to produce dark matter templates. Eventually we plan to involve other astrophysical messengers and observations, especially neutrinos and IceCube data to place more robust constraints on the properties of VHDM particles.

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Appendix

Geometric Derivations

Here we provide a derivation of geometric quantities in Equation 3. Let us start with deriving $l_{\text{max}}(\psi)$, the distance from the Sun to the end of a line of sight in kpc. Consider a sphere of radius R_{mw} (the radius of the Milky Way galaxy) filled with dark matter, centered on the Galactic Center (GC). Starting in Cartesian coordinates centered on the Sun, the sphere satisfies

$$(x - R_{\text{sc}})^2 + y^2 + z^2 = R_{\text{mw}}^2, \tag{16}$$

which accounts for the distance between the Sun and the GC (R_{sc}). We will now describe positions on this sphere using a spherical coordinate system (p, ϕ, θ) like the one in Figure 11.

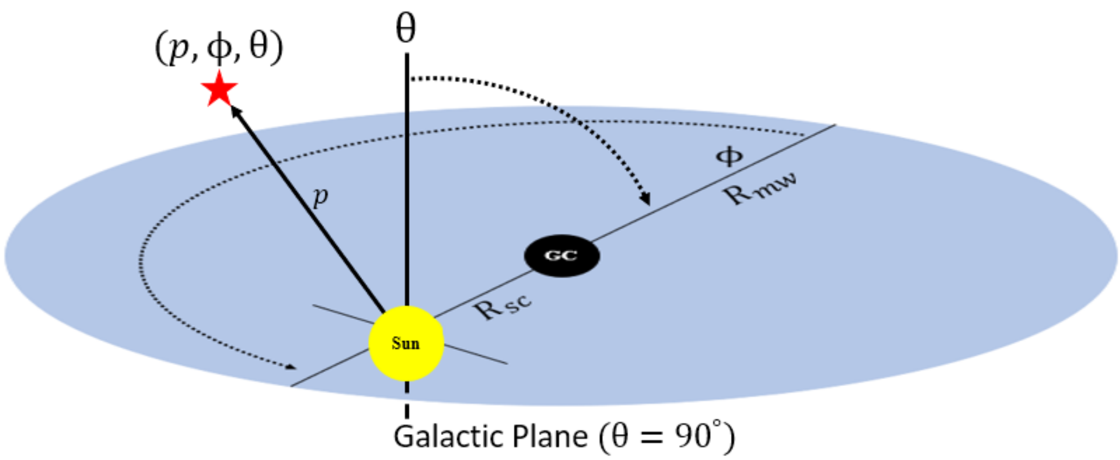

Figure 11: Spherical coordinate system centered on the Sun (not to scale).

Since all lines of sight end on the sphere, we can define the customary conversion between Spherical and Cartesian coordinates in terms of l_{max} as shown in Equation 17.

$$\begin{aligned} x &= l_{\text{max}} \sin(\theta) \cos(\phi) \\ y &= l_{\text{max}} \sin(\theta) \sin(\phi) \\ z &= l_{\text{max}} \cos(\theta) \end{aligned} \tag{17}$$

By substituting relations from Equation 17 into Equation 16, we can express the sphere in terms of the Spherical coordinate system centered on the Sun:

$$l_{\text{max}}^2 + R_{\text{sc}}^2 - 2 l_{\text{max}} R_{\text{sc}} \cos(\phi) \sin(\theta) = R_{\text{mw}}^2. \tag{18}$$

After solving Equation 18 for l_{max}, we are left with

$$l_{\text{max}} = R_{\text{sc}} \cos(\phi) \sin(\theta) + \sqrt{R_{\text{mw}}^2 - R_{\text{sc}}^2 + R_{\text{sc}}^2 \cos^2(\phi) \sin^2(\theta)}. \tag{19}$$

Now, let us find the relationship between Spherical coordinates (ϕ, θ) and Galactic coordinates (ℓ, b) shown in Figure 12.

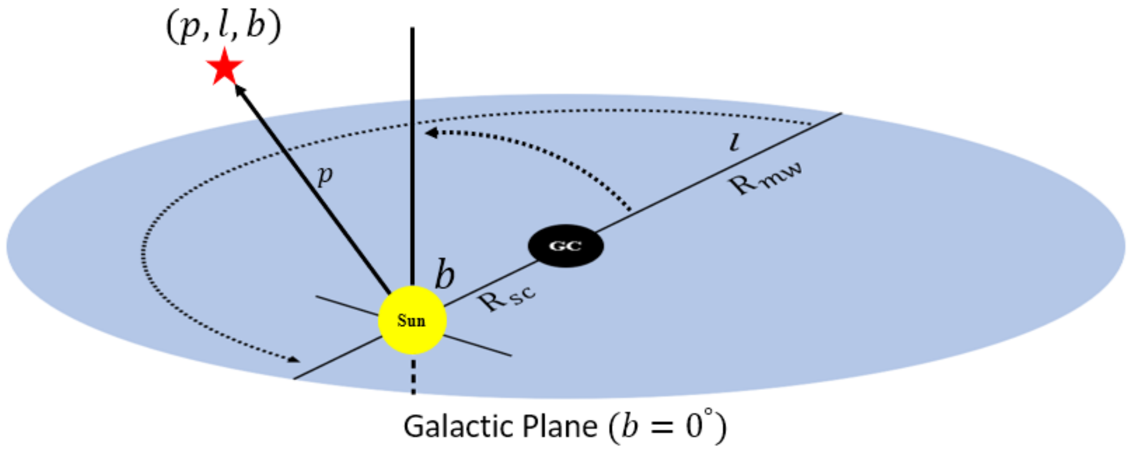

Figure 12: Standard Galactic coordinates system centered on the Sun (not to scale).

Since their direction of increase and origin are the same, ϕ and ℓ are one-to-one. Since θ only opens "down" towards the Galactic Plane (GP) and b starts at the GP and can open either way, they are not one-to-one. The relations are given by

$$\begin{aligned} \phi &= \ell \\ \theta &= \frac{\pi}{2} - b \end{aligned} \quad , \tag{20}$$

which allows us to find l_{max} in terms of Galactic coordinates as given in Equation 21.

$$l_{\text{max}} = R_{\text{sc}} \cos{(b)} \cos{(\ell)} + \sqrt{R_{\text{mw}}^2 - R_{\text{sc}}^2 + R_{\text{sc}}^2 \cos^2{(b)} \cos^2{(\ell)}} \tag{21}$$

We can go a step further and write l_{max} in terms of ψ, the pointing angle with respect to the GC direction. First, let us define a unit vector for a line of sight in Galactic coordinates as

$$\vec{u} = \cos{(\ell)} \cos{(b)}\; \hat{x} + \sin{(\ell)} \cos{(b)}\; \hat{y} + \sin{(b)}\; \hat{z}, \tag{22}$$

and the GC direction as

$$\vec{v} = \hat{x}. \tag{23}$$

Let us define ψ as the angle between $\hat{u}$ and $\hat{v}$. By using the geometric definition of the dot product of $\hat{u}$ and $\hat{v}$,

$$\hat{u} \cdot \hat{v} = ||\hat{u}||\; ||\hat{v}|| \cos{(\psi)}, \tag{24}$$

we find the relation between ψ and (ℓ,b) to be

$$\cos{(\psi)} = \cos{(b)} \cos{(\ell)}, \tag{25}$$

and then use this substitution and the fundamental Pythagorean identity along with Equation 21 to at last arrive at $l_{\text{max}}(\psi)$ in Equation 26.

$$l_{\text{max}} = R_{\text{sc}} \cos(\psi) + \sqrt{R_{\text{mw}}^2 - R_{\text{sc}}^2 \sin^2(\psi)} \tag{26}$$

Now we will derive $r(\psi, l)$, the distance from a point along a line of sight to the GC. The expression for the distance between two vectors (p_1,ϕ_1,θ_1) and (p_2,ϕ_2,θ_2) in Spherical coordinates is given by

$$d = \sqrt{p_1^2 + p_2^2 - 2p_1p_2(\sin(\theta_1)\sin(\theta_2)\cos(\phi_1 - \phi_2) + \cos(\theta_1)\cos(\theta_2))}, \tag{27}$$

which can be adapted for (ℓ,b) by applying the transformations in Equations 20. Let us define Galactic coordinate vectors for the GC and a position on the line of sight respectively:

$$\begin{aligned} \vec{w_1} &= (R_{\text{sc}}, 0, 0) \\ \vec{w_2} &= (l, \ell, b) \end{aligned} \quad . \tag{28}$$

Finally, by using the Galactic coordinate form of Equation 27 on the vectors in Equation 28 and applying the transformation in Equation 25 , the distance from a point along a line of sight to the GC is given by

$$r(\psi, l) = ||\vec{w_2} - \vec{w_1}|| = \sqrt{R_{\text{sc}}^2 + l^2 - 2lR_{\text{sc}}\cos(\psi)}. \tag{29}$$

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